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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

SYRINGE AND NEEDLE HYPODERMIC, Disposable, U-100 Insulin

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers disposable syringes with a permanently attached needle suitable for the administration of U-100 insulin.

Salient Characteristics:

Classification. Syringe sizes, types, and designs are classified as follows:

Sizes	Description
1/2	50 Units (0.5 milliliter)
1	100 Units (1.0 milliliter)

TYPE	Description
I	Single scale
II	Dual scale

Design	Description
A	27 gage needle
B	28 gage needle

Syringe. The syringe shall be transparent. The open end of the syringe shall have an integral flange flattened on two sides for finger tip and anti-roll control. The distal tip of the barrel shall be formed concentric to the longitudinal axis of the barrel. The plunger tip shall have a minimum of two rings making contact with the barrel.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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The distal plunger tip ring shall accurately indicate the syringe fluid volume on the barrel scale. The plunger tip shall be of a color which clearly contrasts with the syringe barrel for maximum visibility.

The plunger tip shall be lubricated with a pharmaceutical grade silicone lubricant. The proximal end of the plunger shall terminate in a flat knob or finger grip. For design B Syringes, the plunger shall be white.

Needle. The needle shall be 27 or 28 gage, 0.5 inch (1.3 cm) ± 0.05 inches in length. The needle shall make a mechanically stable, leakproof connection with the syringe. The needle shall be permanently attached to the syringe. The needle shall have a plastic protective cover which is readily removable, and which does not cause the needle to separate from the syringe when the cover is removed. The longitudinal axis of the needle shall not deviate from that of the syringe by more than 6 degrees 15 minutes.

The cannula shall be coated with a medical grade silicone lubricant. The point shall be sharp to minimize patient trauma. The heel of the needle point shall be sufficiently dulled to prevent coring. There shall be no bent or dull points. The interior and exterior of the cannula shall be free from burrs, slivers, or any foreign matter.

Graduations. The syringe barrel shall have a graduated scale approximately in line with the flattened side of the finger grip flange. The scale shall be readable with the syringe held in the vertical position (i.e., with the needle pointing up). The scale shall be legible and sufficiently dark so that it is visible when viewed through the opposite side of the barrel.

Type I - Single scale graduations. Main graduations shall be 28 percent longer than subgraduations (as a minimum). Main graduations shall be numbered. The word "units" shall appear adjacent to the highest numbered graduation.

The size 1/2 syringe shall have subgraduations every single unit and main graduations every 5 units. The scale shall be 4.3 cm (minimum) in length.

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The size 1 syringe shall have subgraduations every two units and main graduations every 10 units. The scale shall be 5.5 cm (minimum) in length.

Type II - Dual scale graduations. Dual scale syringes shall have the single scale described above, except that main graduations shall be a minimum of 12.5 percent longer than subgraduations. In addition, dual scale syringes shall have an odd-numbered scale adjacent to the even-numbered scale. The main graduations of the odd-numbered scale shall be numbered.

The size 1 odd-numbered scale shall have subgraduations every 2 units and main graduations every 10 units. Subgraduations shall range from 1 to 99 units and main graduations shall range from 5 to 95 units.

Dual scale syringes may be substituted for single scale syringes only if all single scale requirements (including the 33 percent longer main graduations) are met.

Markings. Syringes shall bear the legend "For Use With U-100 Insulin Only" or words to that effect. Syringes shall bear the manufacturer's name or registered trademark.

Dead Space. Dead space shall not exceed one (1) unit in the syringe and needle.

Materials. The syringe barrel and plunger shall be manufactured from polypropylene plastic. As an alternate, the plunger may be manufactured from impact-modified polystyrene have having an impact strength (Izod) of 0.5 ft. - lbs/inch of notch (min) when tested in accordance with ASTM Designation D256. The plunger tip shall be manufactured from rubber. The needle shall be manufactured from austenitic Class 3 stainless steel in accordance with ASTM Designation F899. The syringe shall be properly annealed and passivated.

Toxicity, pyrogenicity, and sterility. Shall be non-toxic and pyrogenic in accordance with USP. Shall be sterile in accordance with USP or AAMI guidelines.

Workmanship. The syringe and needle shall contain no defects which detract from appearance or impair the serviceability of the unit.

Instructions. Each unit package shall contain complete instructions for use and disposal of the syringes.

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PRODUCT QUALITY LEVEL

Visual and dimensional examination. Shall be examined to verify that the design, construction and workmanship are in accordance with applicable requirements. Needle keenness examination shall be at 10x magnification.

Materials. The supplier shall be responsible for performing tests to assure that materials are in accordance with requirements as specified.

Unit. Package (PG). One package containing one hundred (100) syringe and needle hypodermic of one size and type only packed in a readily opened container so constructed as to protect the needle cannula and maintain the sterility of the syringe and needle combination until manually opened.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for examination purposes shall be one syringe and needle. Sampling shall be inspection level II with an acceptable quality level (AQL) of 6.5, except the AQL for needle keenness. (See Note 3).

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Examination. Examination shall be conducted to determine compliance with specification requirements.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for test purposes shall be one syringe and needle. Sampling for Leakage, Accuracy and Resistance to separation shall be S-1 with an AQL of 1.0. (See Note 3).

Accuracy of graduations. The graduated scale shall be accurate to within ± 5 percent of the volume indicated at volumes equal to or greater than 0.2 ml. The scale shall be accurate to within 0.007 ml volume less than 0.2 ml. Syringe shall be tested for volumetric accuracy using a calibrated burette.

Resistance to separation. Syringe and needles shall be tested for resistance to separation. The needle shall not separate from the syringe when the needle is subjected to a 5 pound tensile force for one minute. (See Note 2).

Leakage. An amount of water equal to more than 1/2 the capacity shall be drawn into the syringe through the hypodermic needle. The syringe shall then be held vertically (needle up) to allow air to rise above the water and be expelled. Tapping the barrel of the syringe will facilitate removal of trapped air. The plunger shall than be pressed inward to the 50% mark. Either crimp the needle or block the needle with rubber plug so no water may leak from the needle. Apply a 1 (one) pound axial air force for size 1 syringes. Apply the force for 10 seconds minimum. Observe the cannula/hub junction and barrel/hub interface for possible leakage. No water shall pass the cannula/hub junction and barrel/hub interface. No water shall pass the front rib of the plunger tip.

Sterility and pyrogenicity. Syringe and needle units shall be tested for freedom from pyrogens in accordance with the USP. Using either the Bacterial Endotoxin Test or the Pyrogen Test. Sterility testing shall be in accordance with USP or AAMI guidelines.

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Biocompatibility. Plastic material shall be tested in accordance with the USP tests for Class IV plastics. Rubber material shall be tested in accordance with the Intramuscular and Intracutaneous sensitivity tests of the USP.

Certificates of Biocompatibility. A certificate of compliance, stipulating, that offered items meet the biocompatibility test, shall be acceptable, provided that the contractor has supplied acceptable items to the Government within the past three years, and that there have been no changes to material, formulation, and method of processing since actual testing was performed. The certificate shall include the date of actual testing, the test method and results, the name and title of the person evaluating results, the current contract number and of the person evaluating results, the current contractor number and NSN, a statement to the effect that material. processing, and formulation have not changed since actual testing was performed, certification.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers'own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and /or order.

NOTE (1): The following National Stock Numbers, their respective Item Identifications, Types, Designs, and Sizes are covered by this document:

NSN	ITEM IDENTIFICATION	SPECIFICATION TYPE	DESIGN	SIZE
6515-00-597-8070	SYRINGE AND NEEDLE, Disposable, U-100 Insulin, 1 ml. 27 or 28 gage needle	II	A or B	1
6515-01-058-5953	SYRINGE AND NEEDLE, Disposable, U-100 Insulin, 1 ml, 27 gage needle	I	A	1

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6515-01-125-6606	SYRINGE AND NEEDLE, Disposable, U-100 Insulin, 1/2 ml, 27 gage needle	I	A	1/2
6515-01-172-1090	SYRINGE AND NEEDLE, Disposable, U-100 Insulin, 1 ml, 28 gage needle	I	B	1
6515-01-089-1069	SYRINGE AND NEEDLE, Disposable, U-100 Insulin, 1/2 ml, 28 gage needle	I	B	1/2

NOTE (2): An alternate test, such as a pull-to-destruction test, is acceptable in lieu of the 5 pound separation test provided product found acceptable of the pull-to-destruction test also passes the 5 pound separation test.

NOTE (3): As an alternate, a statistically equivalent sampling plan which would ensure a comparable degree of freedom from defects is acceptable.

Ordering data (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4991

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